# Ticket 1

TCK01-PwReset

## Summary

| | |
|---|---|
| Category | Account management |
| Requester | Bob Nasser (Sales) |
| Request | Forgot password, cannot sign in to the workstation |
| Action | Reset to a temporary password and force a change at next logon |
| Tools | Active Directory Users and Computers (ADUC), PowerShell |
| Outcome | Resolved. User signed in with the temporary password and set a password only he knows |

## Problem

Bob Nasser (Sales) forgot his password and cannot sign in to the workstation. Before changing anything, the requester's identity was verified with their employee ID. A password reset hands over account access, so an unverified request is a social-engineering risk. The verification method is recorded in the ticket before the reset is made.

## Environment

| Item | Detail |
|---|---|
| Domain | acme.local |
| Account | bnasser (Bob Nasser), OU: Acme → Users → Sales |
| Tools | Active Directory Users and Computers (ADUC), PowerShell |
| Client | Domain-joined Windows client VM (sign in to: ACME) |

## Troubleshooting Steps

| # | Action | Result |
|---|---|---|
| 1 | Verify the requester with their employee ID before changing anything. | Identity verified; method recorded in the ticket before the reset. |
| 2 | Open ADUC with `dsa.msc` (Win+R). | ADUC opens. |
| 3 | Locate the user: Acme → Users → Sales → Bob Nasser. | Account found. |
| 4 | Right-click the user and choose Reset Password... | Reset Password dialog opens. |
| 5 | Set a temporary password and tick **User must change password at next logon**. Leave **Unlock the user's account** unticked (a reset does not unlock an account, and this account was not locked). | Directory confirms the password has been changed. |
| 6 | Client: sign in with the old password. | Rejected: user name or password is incorrect. |
| 7 | Client: sign in with the temporary password. | Accepted; Windows forces a password change. |
| 8 | Client: user enters old password, new password and confirmation (hidden from support). | Your password has been changed. |
| 9 | Client: first sign-in. | Windows builds a new local profile (takes a few minutes the first time); desktop loads. |

| # | Action | Result |
|---|---|---|
| 10 | Server: run `Get-ADUser bnasser -Properties PasswordLastSet,PasswordExpired` before and after the first sign-in. | Before: blank / True. After: 10/1/2026 3:52:33 AM / False. |
| 11 | Edge case: the user's new password is rejected (tried to reuse the same password). | Blocked by the password history rule; see Lessons Learned. |

## Root Cause

The user forgot his password. There was no fault on the account itself. Note: a reset is not an unlock. If sign-in still fails after a reset, check whether the account is locked or disabled (see Ticket 02).

## Fix

Reset the password in ADUC to a temporary password with **User must change password at next logon** ticked. This setting is the point of the ticket: it means only the user ever knows his real password. **Unlock the user's account** was left unticked because a reset does not unlock an account, and this account was not locked.

**Note on the temporary password:** the lab uses a simple temporary password for demonstration. In production it should be randomly generated, unique per ticket, and passed to the user over a secure channel (phone or approved portal, never plain email or chat).

**PowerShell equivalent**

```
# Reset to a temporary password (prompts, so it never appears in history)
Set-ADAccountPassword -Identity bnasser -Reset -NewPassword (Read-Host -AsSecureString "Temp password")

# Force a change at next logon
Set-ADUser -Identity bnasser -ChangePasswordAtLogon $true
```

## Verification

**From the client:** the old password no longer works; the temporary password works and Windows forces a change; the user chooses a new password (old, new, confirmation), which is hidden so nobody on the support side sees it; first sign-in builds a new local profile and reaches the desktop.

**In Active Directory:** the server-side check proves the "must change" flag was set and then cleared by the user, not by support:

```
Get-ADUser bnasser -Properties PasswordLastSet,PasswordExpired |
  Select-Object Name,PasswordLastSet,PasswordExpired
```

| Run | PasswordLastSet | PasswordExpired | Meaning |
|---|---|---|---|
| After the reset, before first sign-in | *(blank)* | `True` | Flag set; user has not chosen a password yet |
| After the user signed in and changed it | `10/1/2026 3:52:33 AM` | `False` | User set his own password |

**Why PasswordLastSet is blank in the first run:** ticking "must change at next logon" sets the account's `pwdLastSet` attribute to `0`. AD reads that as "password already expired", and PowerShell displays it as empty. Once the user changes the password, AD writes a real timestamp.

## Lessons Learned

• **Nobody can read a password back.** AD stores only a hash. Support can overwrite a password with a new reset, which is why every reset is logged in the ticket with who asked and how they were verified.

• **A reset is not an unlock.** If sign-in still fails after a reset, check whether the account is locked or disabled (see Ticket 02).

• **When the user's new password is rejected:** Windows shows a generic error covering length, complexity and history. Here the user tried to reuse the same password, which the **password history** rule blocks. The technician has to work out which rule applies:

| Cause | What to tell the user |
|---|---|
| Too short | Use at least the minimum length |
| Not complex enough | Mix upper case, lower case, numbers, and symbols; avoid the user name |
| Reused a recent password | Choose something not used before, including the temporary password |

## Screenshots

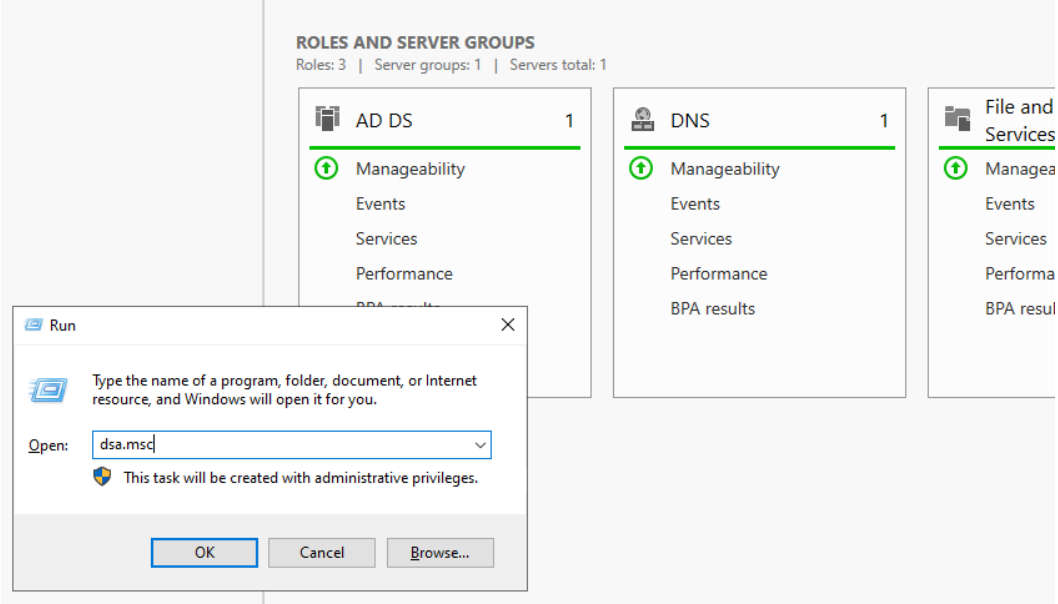



*Figure 1. Run dialog with dsa.msc*

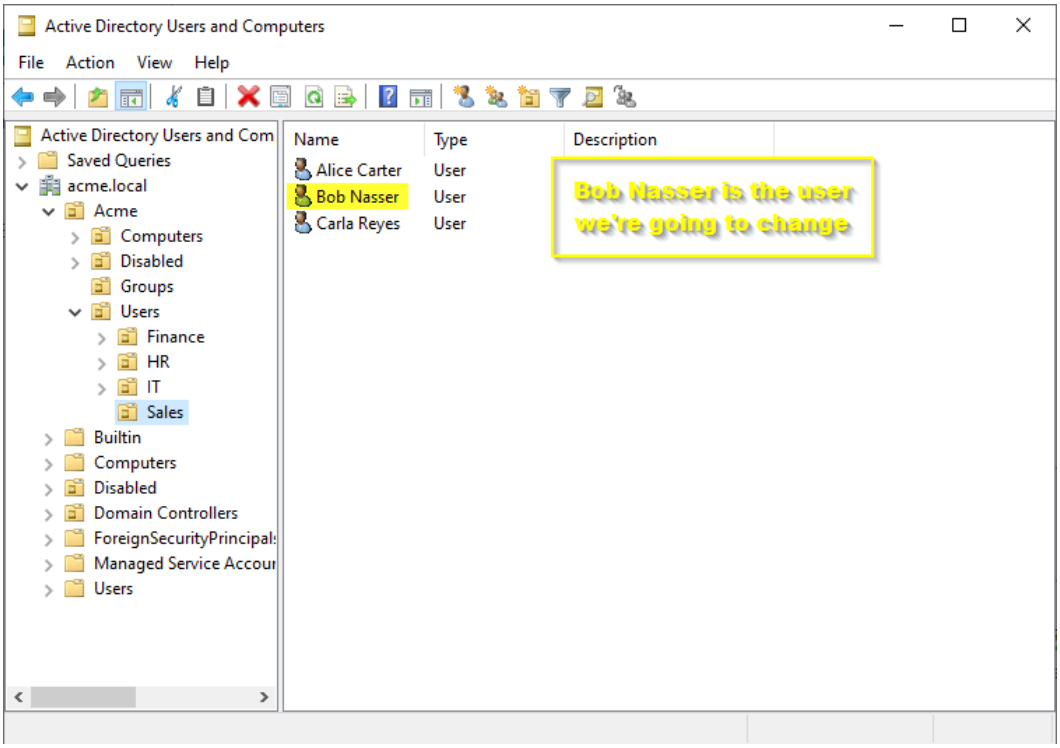



*Figure 2. ADUC with Bob Nasser selected in the Sales OU*

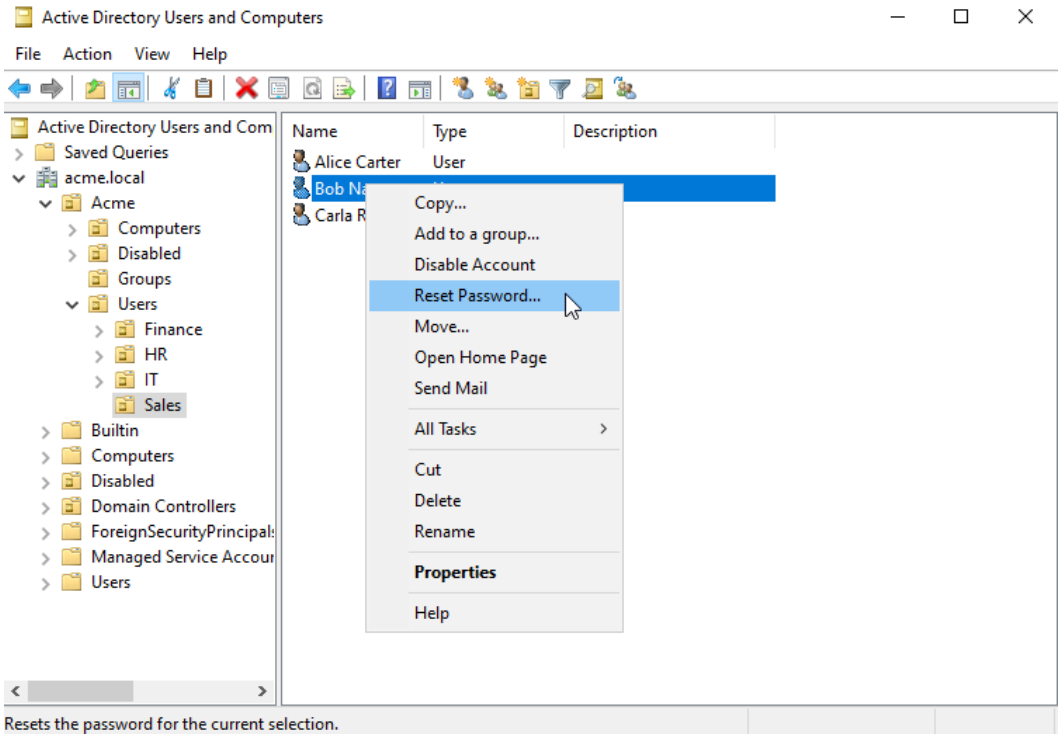



*Figure 3. Context menu with Reset Password highlighted*

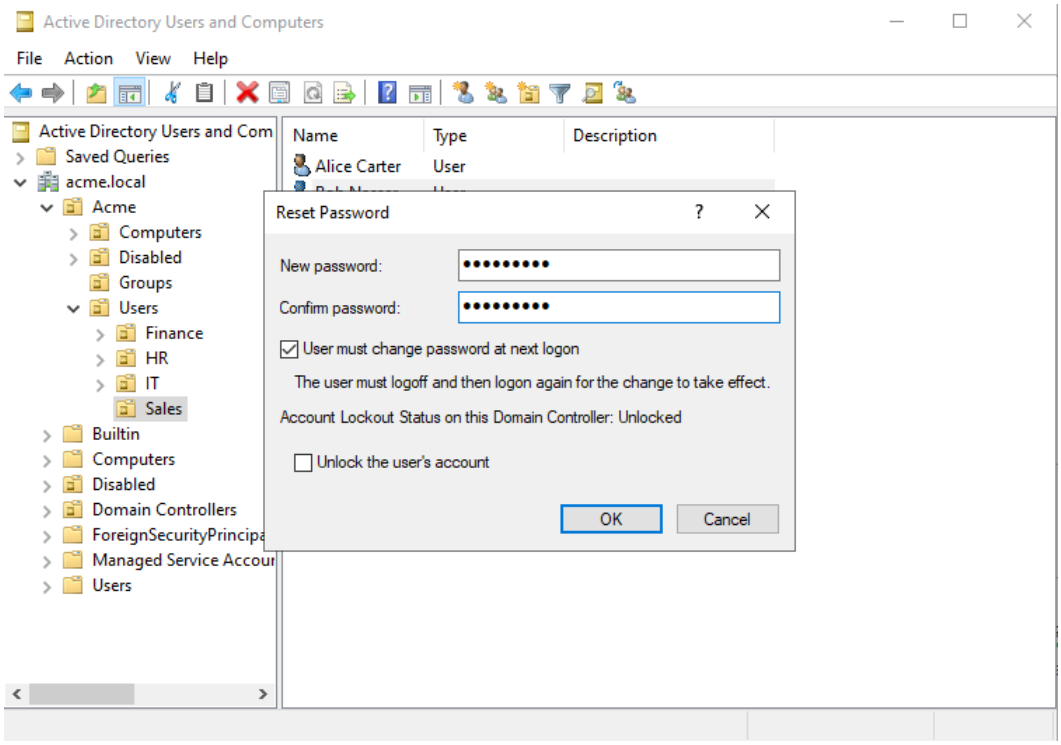



*Figure 4. Reset Password dialog with the must-change option ticked*

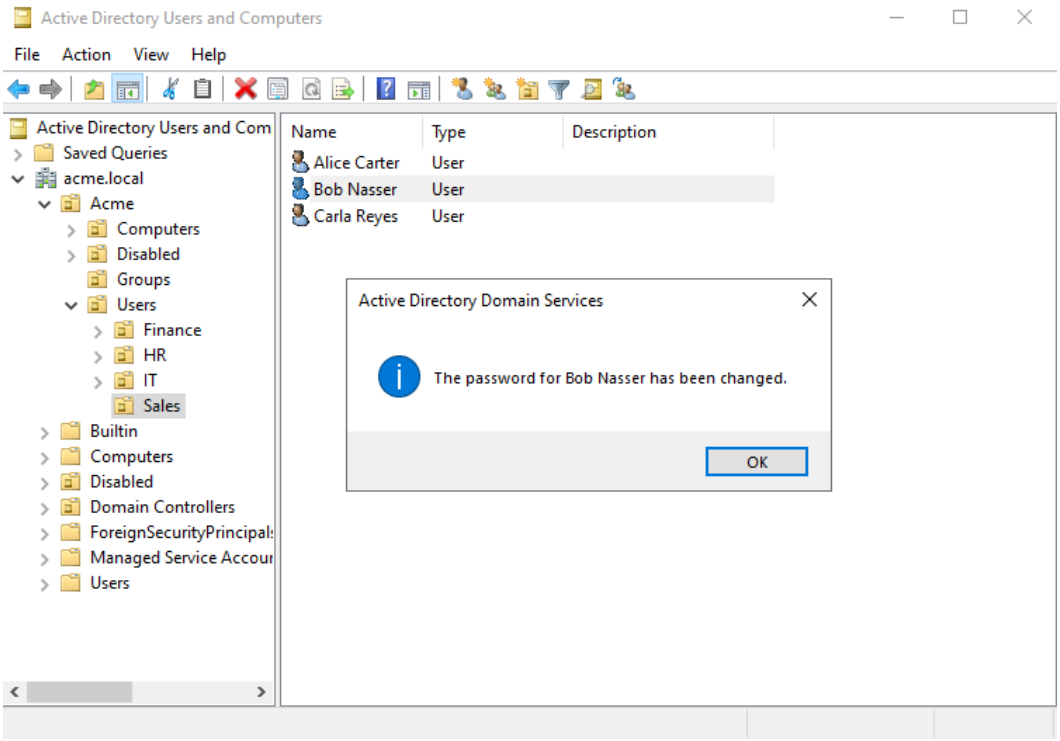



*Figure 5. Confirmation that the password has been changed*

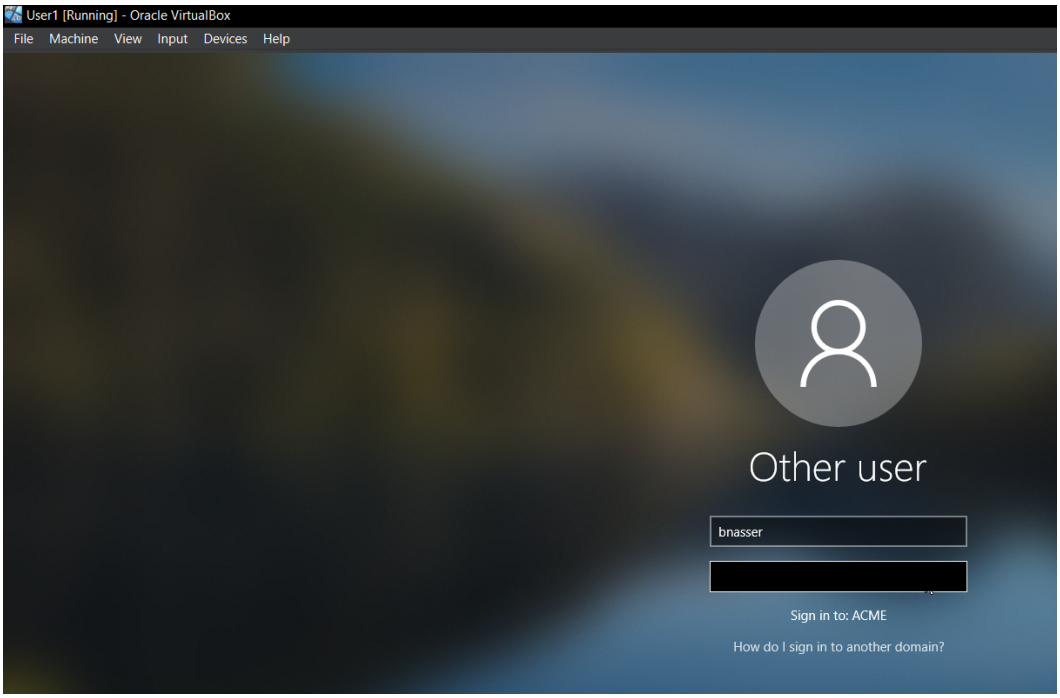



*Figure 6. Sign-in attempt with the old password*

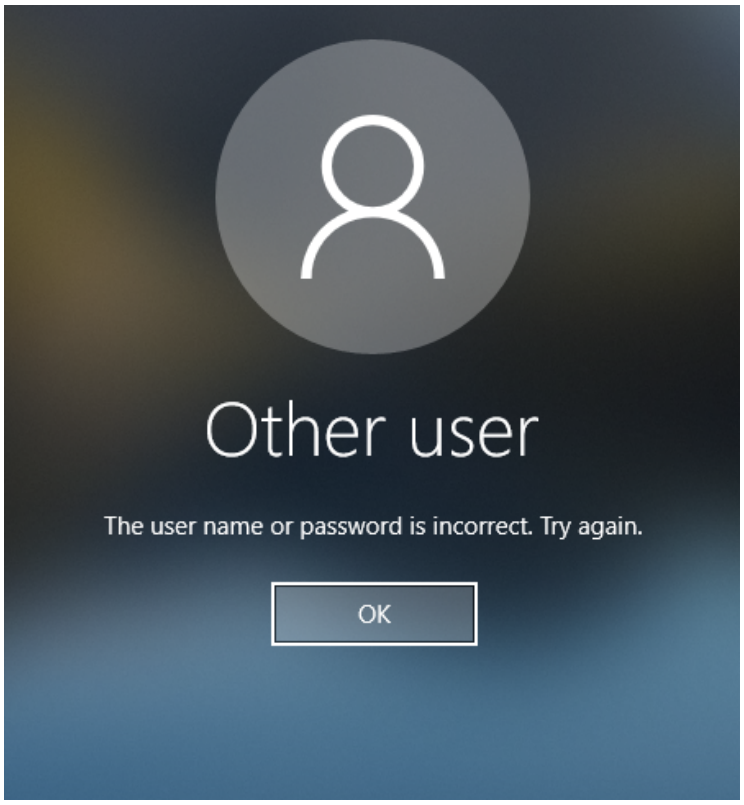



*Figure 7. Sign-in error: user name or password is incorrect*

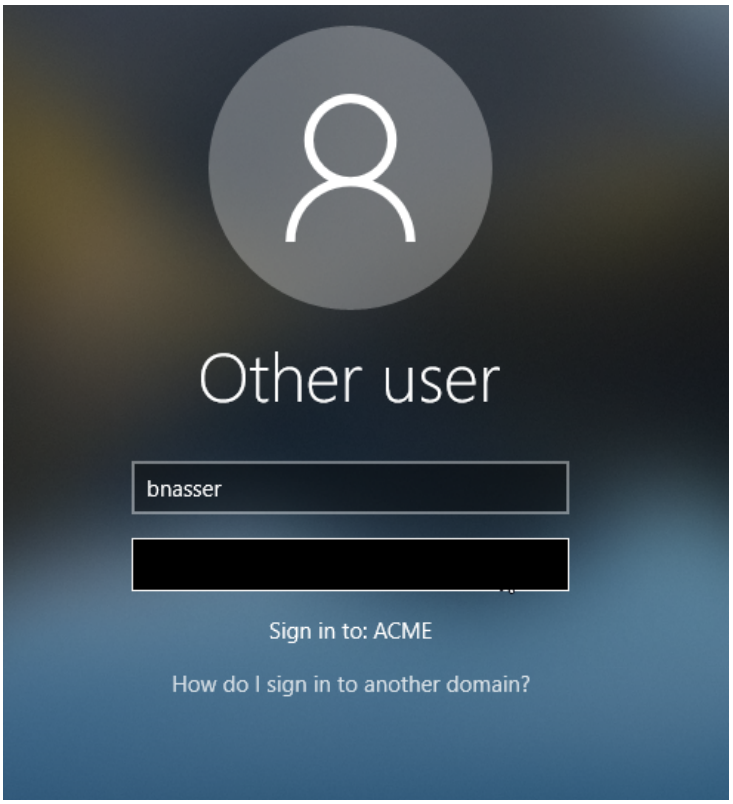



*Figure 8. Sign-in with the temporary password*

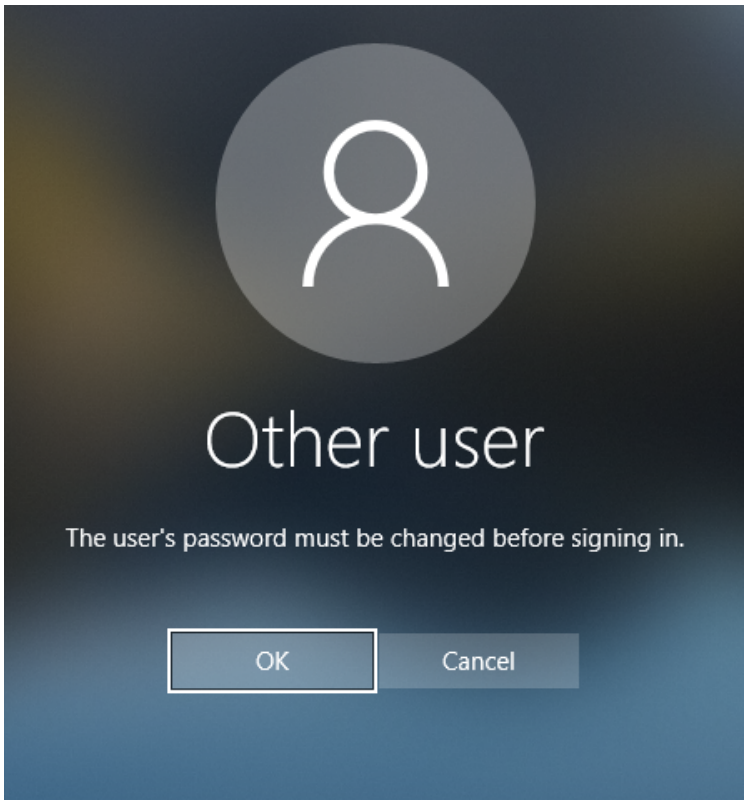



*Figure 9. Prompt: password must be changed before signing in*

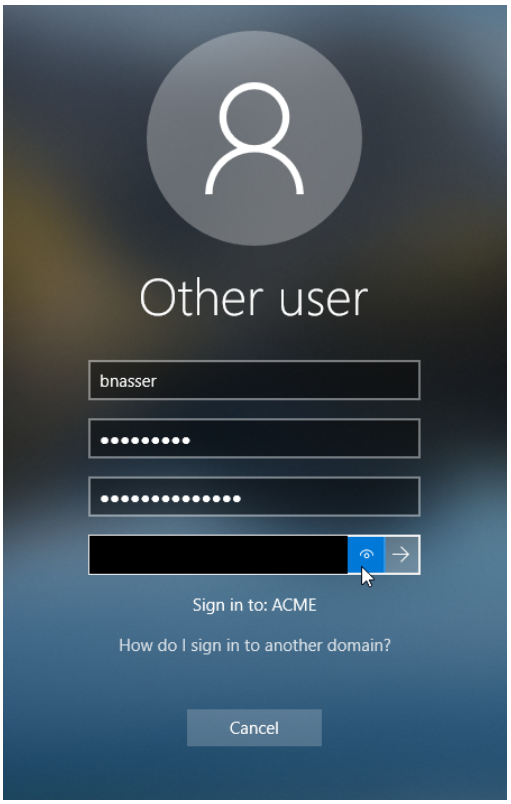



*Figure 10. Change password screen*

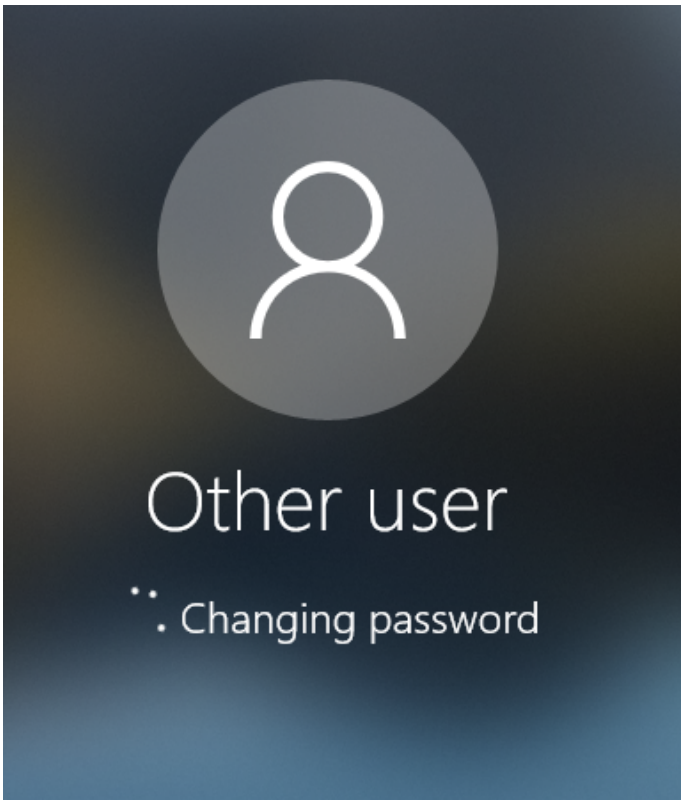



*Figure 11. Changing password progress*

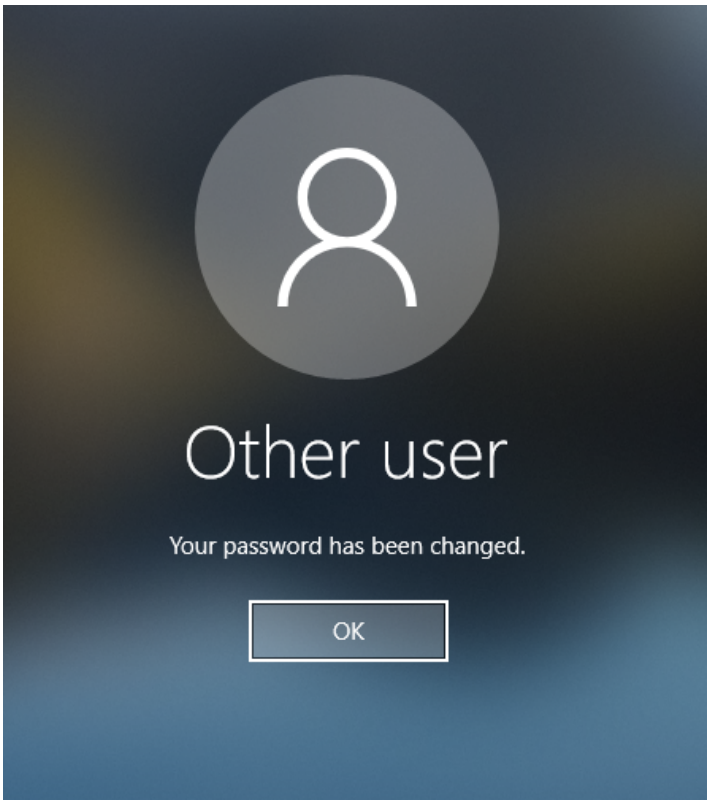



*Figure 12. Your password has been changed*

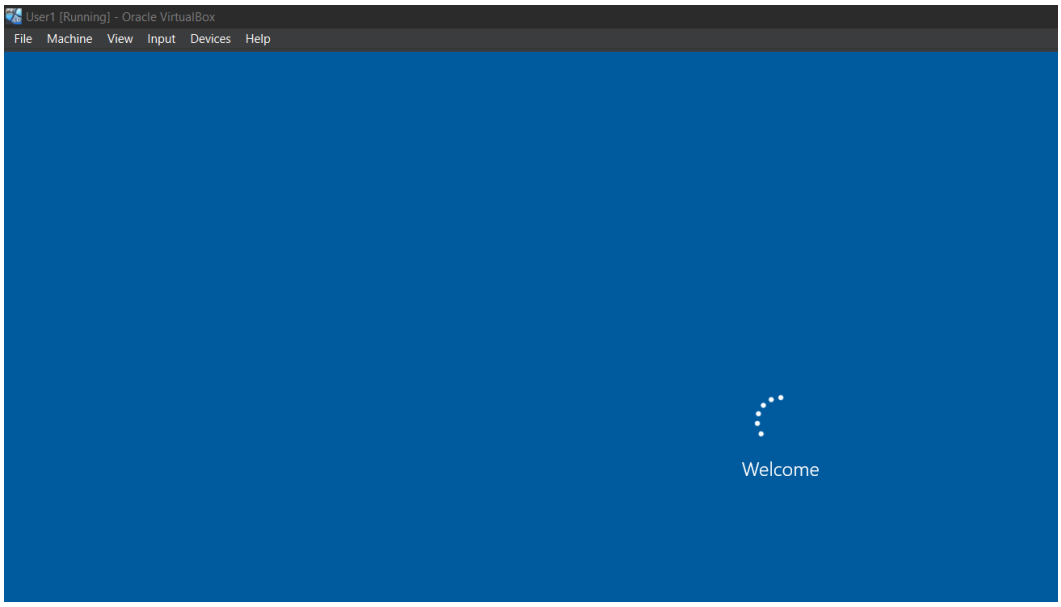



*Figure 13. Welcome screen*

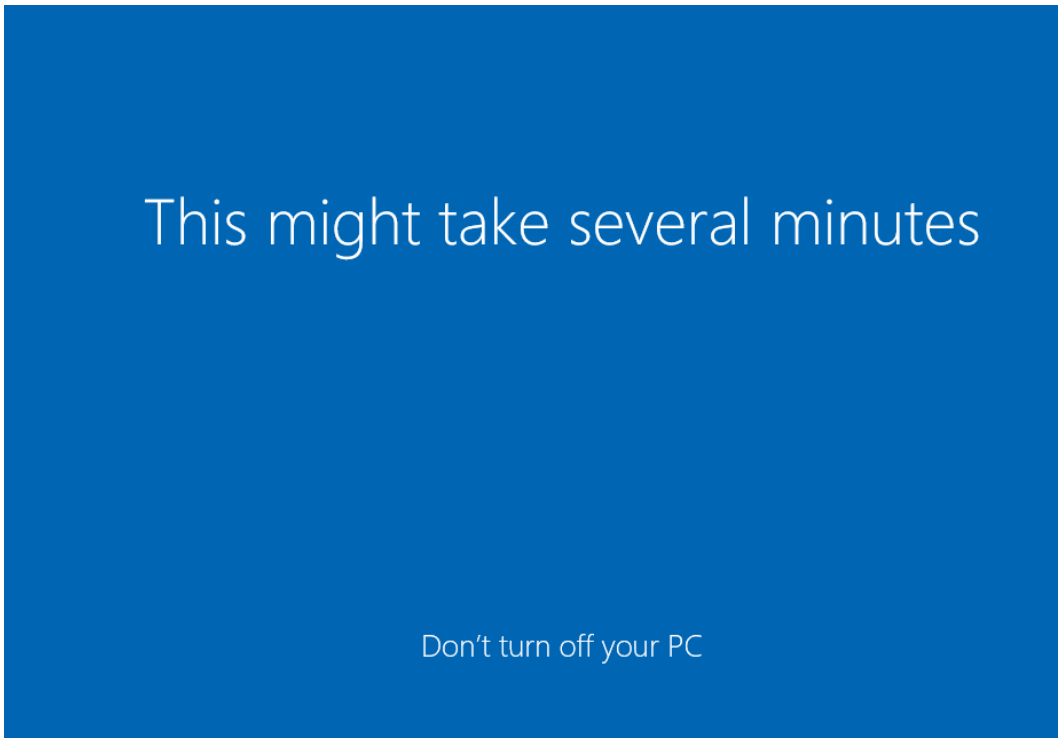



*Figure 14. First-time sign-in setup*

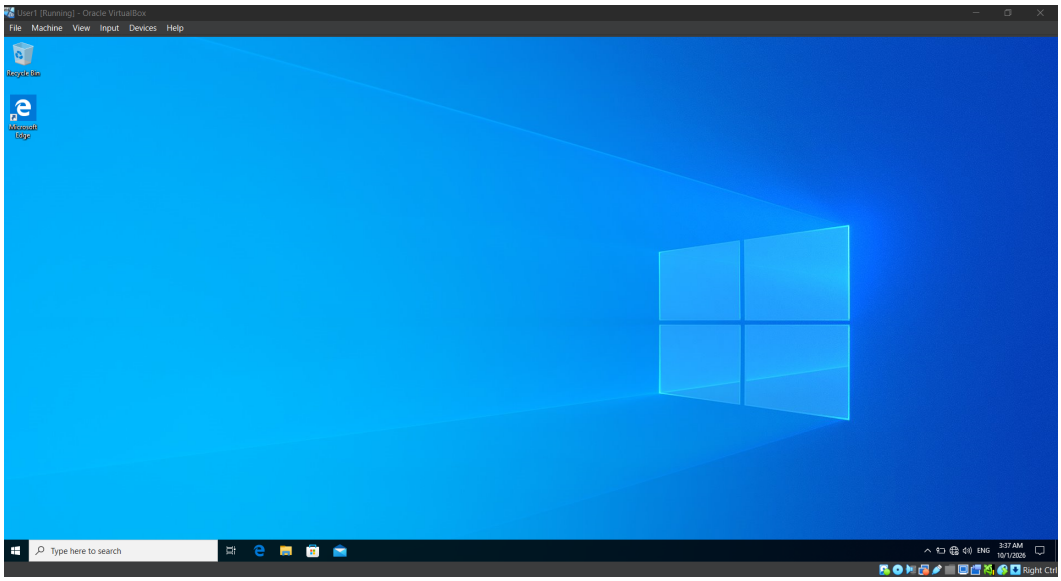



*Figure 15. Desktop after successful sign-in*

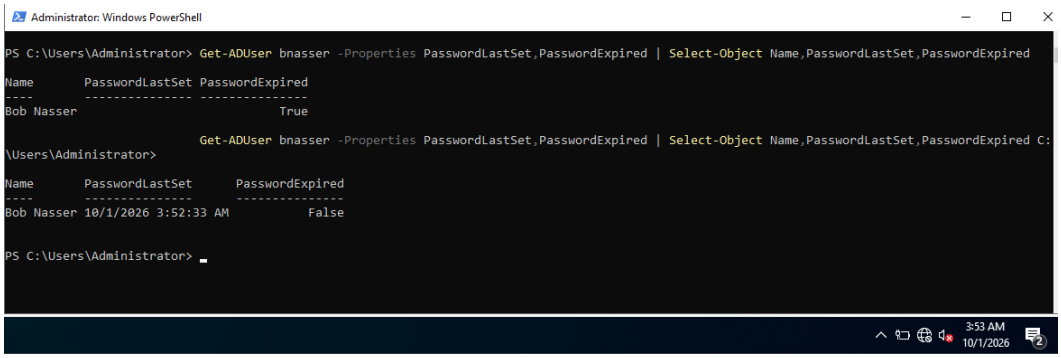



*Figure 16. PowerShell output showing both runs*

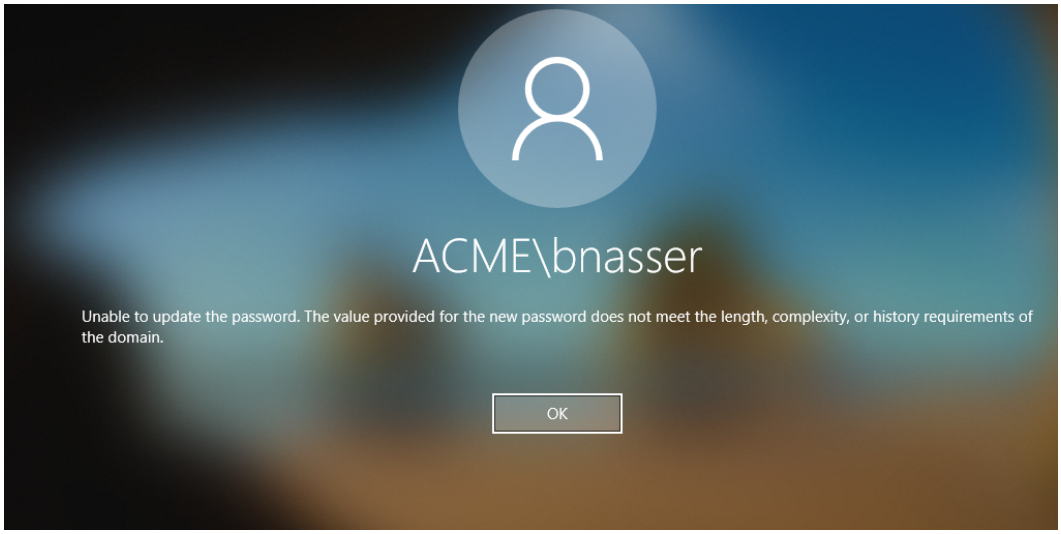



*Figure 17. Error: password does not meet length, complexity, or history requirements*